

Homework 8 (Heston model)

$$\begin{cases} dS_t = rS_t dt + \sqrt{v_t} S_t dW_t^S \\ dv_t = \kappa(\theta - v_t)dt + \sigma\sqrt{v_t}dW_t^v \end{cases}$$

$$S_0 = 100, r = 0.02, T = 1$$

Heston parameters:

$$\kappa = 2.0, \theta = 0.04, \sigma = 0.5, \rho = -0.7, v_0 = 0.04$$

Simulation:

- Time steps: $N = 250$
- Paths: $M = 10,000$

Task 1)

- Simulate the processes (S, v)
- Plot 10 sample paths of S_t and 10 sample paths of v_t

Task 2)

- Simulate Black–Scholes with: $\sigma = \sqrt{\theta} = 20\%$
- Compare terminal distributions of S_T :
 - mean
 - variance
 - skewness
 - kurtosis

Task 3)

- Price a European call with $K = 90, 100, 110$
- Compare Heston price vs Black–Scholes price

Task 4)

- Compute call prices under Heston for several $K \in [70, 130]$
- For each strike invert Black–Scholes to obtain implied volatility
- Plot the implied volatility $IV(K)$

Task 5)

- Set: $\rho = -0.9, -0.5, 0$

- Recompute smile
- Plot all curves together
- Idem for $\sigma = 0.2, 0.5, 1.0$

Task 6)

(Comment: with base parameters: $2 \cdot 2 \cdot 0.04 = 0.16 < 0.25$ Feller condition is violated)

- Simulate variance paths
- Check if variance becomes negative
- Apply “full truncation”: $\sqrt{v_t} \rightarrow \sqrt{\max(v_t, 0)}$

Task 7)

- Implement the characteristic function of log-price: $\phi(u) = \mathbb{E}[e^{iu \log S_T}]$ under the Heston model.
- Implement numerical integration (e.g. Simpson or trapezoidal) for the price of the call
- Compute price for:

$$K = 80, 90, 100, 110, 120$$

- Compute the same option prices using Monte Carlo
- Compare: C_{Fourier} vs C_{MC}
- Report absolute error and comment on efficiency and accuracy

DELIVERY: PDF report + (runnable) code/notebook with software at your convenience

(R/Python/C#/VBA..)

DEADLINE: May 16 Reports submitted earlier will receive a higher bonus